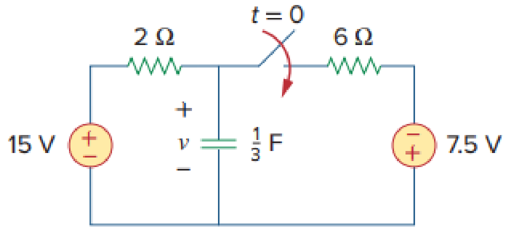
Chapter 7, Problem 10PP

Problem

Find $v(t)$ for $t > 0$ in the circuit of Fig. 7.44. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5$.

Fig. 7.44:



Step-by-step solution

Step 1 of 10

Refer Figure 7.44 from the textbook.

For $t < 0$, the switch is opened for a long time and the capacitor acts like as open circuit to dc.

Draw the circuit for $t < 0$ as shown in Figure 1.

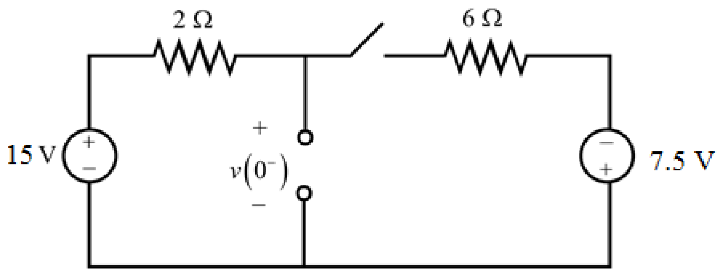


Figure 1

Step 2 of 10

Draw the modified circuit for the circuit in Figure 2.

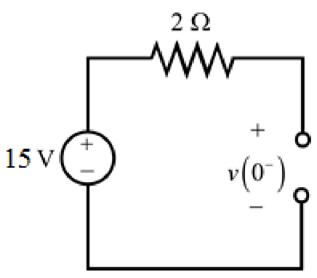


Figure 2

Step 3 of 10

From Figure 2, the current flowing through the circuit is zero. So, the total voltage of 10 V is dropped across the voltage, $v(0^-)$.

$$v(0^-) = 15\text{ V}$$

In capacitor voltage cannot change instantaneously.

$$\begin{aligned} v(0) &= v(0^+) \\ &= v(0^-) \\ &= 15\text{ V} \end{aligned}$$

If the switch is closed for a long time, the circuit reaches steady state and the capacitor acts like an open circuit again.

Step 4 of 10

For $t > 0$, the switch is closed. The circuit in Figure 1 is modified as shown in Figure 3.

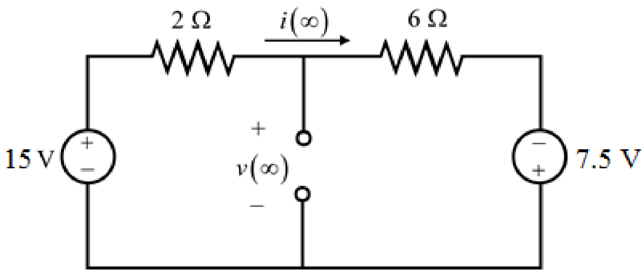


Figure 3

Step 5 of 10

Apply Kirchhoff's voltage law for the circuit in Figure 3.

$$-15+[i(\infty)]2+[i(\infty)]6-7.5=0$$
$$[i(\infty)]8=22.5$$
$$i(\infty)=\frac{22.5}{8}\text{ A}$$

Calculate the voltage, $v(\infty)$.

Apply Kirchhoff's voltage law for the circuit in Figure 3.

$$v(\infty)-15+2[i(\infty)]=0$$
$$v(\infty)=15-2[i(\infty)]$$

Substitute $\frac{22.5}{8}\text{ A}$ for $i(\infty)$ in $v(\infty)=10-2[i(\infty)]$.

$$\begin{aligned}v(\infty)&=15-2\left(\frac{22.5}{8}\right)\\&=15-5.625\\&=9.375\text{ V}\end{aligned}$$

Calculate the equivalent resistance, R_{Th} by shorting the voltage sources. R_{Th}, is the resistance seen through the capacitor. Draw the modified circuit as shown in Figure 4.

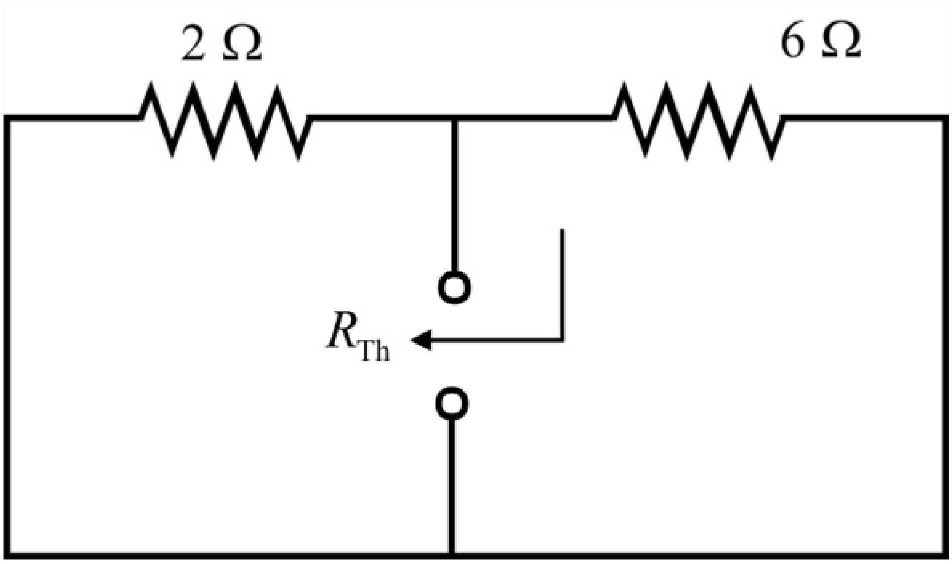


Figure 4

Step 6 of 10

The resistances $2\ \Omega$ and $6\ \Omega$ are connected in parallel. So, the resultant resistance is,

$$\begin{aligned}R_{\text{Th}}&=6\parallel 2\\&=\frac{(6)(2)}{6+2}\\&=\frac{12}{8}\\&=1.5\ \Omega\end{aligned}$$

Step 7 of 10

Calculate the time constant, τ.

$$\tau = R_{Th}C$$

Substitute $1.5\ \Omega$ for R_{Th} and $\frac{1}{3}$ F for C in $\tau = R_{Th}C$.

$$\begin{aligned}\tau &= (1.5)\frac{1}{3}\\ &= 0.5\ \text{s}\end{aligned}$$

Step 8 of 10

Calculate the capacitor voltage, $v(t)$.

$$v(t) = v(\infty) + \left[v(0) - v(\infty)\right]e^{-\frac{t}{\tau}}$$

Substitute $9.375\ \text{V}$ for $v(\infty)$, $0.5\ \text{s}$ for τ and $15\ \text{V}$ for $v(0)$ in $v(t)$.

$$\begin{aligned}v(t) &= 9.375 + (15 - 9.375)e^{\frac{-t}{0.5}}\\ &= 9.375 + 5.625e^{-2t}\ \text{V}\end{aligned}$$

Therefore, the voltage across the capacitor, $v(t)$ for $t > 0$ is $\boxed{9.375 + 5.625e^{-2t}\ \text{V}}$.

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Calculate the voltage, $v(t)$ at $t = 0.5$.

$$\begin{aligned}v(0.5) &= 9.375 + 5.625e^{-2(0.5)}\\ &= 9.375 + 5.625e^{-1}\\ &= 9.375 + 2.0693\\ &= 11.444\ \text{V}\end{aligned}$$

Therefore, the voltage, $v(t)$ at $t = 0.5$ is $\boxed{11.444\ \text{V}}$.

Step 10 of 10

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message Not Found

description The requested resource is not available.

Apache Tomcat/6.0.44